

BIOLOGY
(SCIENCE PAPER – 3)

Maximum Marks: 80

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. *The time given at the head of this Paper is the time allowed for writing the answers.*
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5. *Section A is compulsory. Attempt any four questions from Section B.*
 6. *The intended marks for questions or parts of questions are given in brackets [].*

Instruction for the Supervising Examiner

*Kindly read aloud the Instructions given above to all the candidates present in the
Examination Hall.*

This paper consists of 15 printed pages and 1 blank page.

SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

Select the correct answers to the questions from the given options.

[15]

(Do not copy the questions, write the correct answer only).

- (i) The surgical method of sterilisation in a male involves ligation of the:
 - (a) Oviduct
 - (b) Sperm Duct
 - (c) Testis
 - (d) Uterus
- (ii) Which one of the following does NOT affect the rate of photosynthesis?
 - (a) Oxygen
 - (b) Carbon dioxide
 - (c) Water
 - (d) Temperature
- (iii) A gland which is both endocrine and exocrine in function is:
 - (a) Pituitary
 - (b) Thyroid
 - (c) Pancreas
 - (d) Adrenal

- (iv) What is one way women can contribute to controlling population explosion in India?
- (a) By having more children to contribute to the workforce.
 - (b) By making informed choices about family planning methods.
 - (c) By ignoring health care services.
 - (d) By relying on men to make decisions about family size.
- (v) A 60-year-old man met with an accident and had difficulty in hearing. It was due to a damage to the sensory cells in the cochlea. Which structure has been affected?
- (a) Auditory canal
 - (b) Vestibule
 - (c) Ear drum
 - (d) Organ of Corti
- (vi) **Assertion (A):** Plants growing in shade tend to have larger leaves.
- Reason (R):** This is because they need to capture more sunlight for photosynthesis.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
 - (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(vii) Sonu's grandmother keeps mangoes in a rice bin to ripen. The mangoes ripen after a few days because of the phytohormone:

- (a) Auxin
- (b) Ethylene
- (c) Absciscic acid
- (d) Gibberellin

(viii) **Assertion (A):** The loop of Henle helps concentrate or dilute urine.

Reason (R): The loop of Henle regulates the amount of water, reabsorbed into the bloodstream.

- (a) (A) is true and (R) is false.
- (b) (A) is false and (R) is true.
- (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(ix) According to Charles Darwin's theory of evolution, what drives the process of natural selection?

- (a) Genetic mutations
- (b) Human intervention
- (c) Survival of the fittest
- (d) Random chance

- (x) During atrial systole, the tricuspid valve (P) _____ and semilunar valve (Q) _____.
- (a) closes, opens
 - (b) closes, closes
 - (c) opens, closes
 - (d) opens, opens
- (xi) Some raisins were added to a rice pudding. After a while, it was observed that the raisins had swollen up. What is the scientific phenomenon behind the swelling of these raisins?
- (a) Isotonic, Endosmosis
 - (b) Hypotonic, Exosmosis
 - (c) Hypertonic, Exosmosis
 - (d) Hypotonic, Endosmosis
- (xii) In a dividing cell, the nuclear membrane reappears during:
- (a) Prophase
 - (b) Metaphase
 - (c) Anaphase
 - (d) Telophase

(xiii) A person under the influence of alcohol has difficulty in maintaining body balance and coordinating movements while walking. Which part of the brain is affected in such cases?

- (a) Cerebellum
- (b) Cerebrum
- (c) Medulla oblongata
- (d) Thalamus

(xiv) Surya's father was diagnosed with Diabetes Insipidus. What are the primary symptoms of Diabetes Insipidus that Surya's father must have experienced?

P: Weight gain

Q: More-urine formation

R: Joint pain

S: Excessive thirst

- (a) P and Q
- (b) Q and R
- (c) Q and S
- (d) P and R

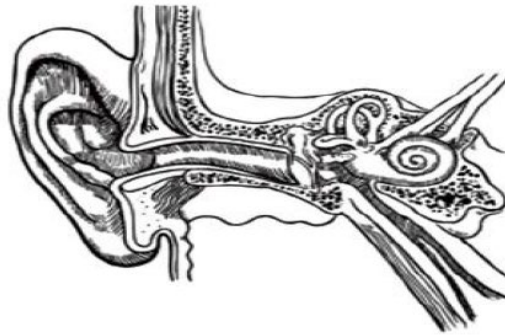
(xv) **Assertion (A):** Transpiration decreases the amount of moisture in the atmosphere.

Reason (R): A dense forest contributes to bringing rain due to transpiration.

- (a) (A) is true and (R) is false.
- (b) (A) is false and (R) is true.
- (c) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

Question 2

- (i) Give the biological / technical terms for the following: [5]
- (a) The respiratory pigment found in RBCs.
 - (b) The veins which start with capillaries and end in capillaries.
 - (c) The part of the eye grafted during eye donation.
 - (d) Collective term for the protective coverings of the brain and spinal cord.
 - (e) The epidermal pores through which guttation takes place.
- (ii) Given below is the transverse section of the human ear. Read the information [5]
below the diagram and fill in the blanks:



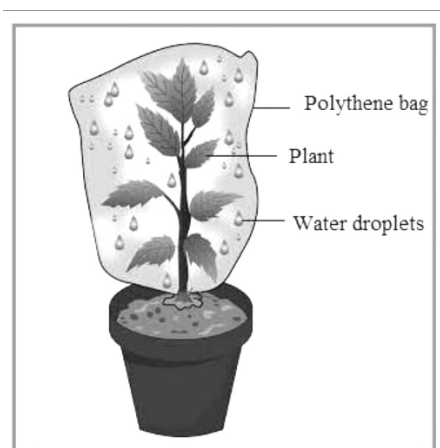
The sense organs enable us to be aware of the conditions of our external and internal environment. The human ear is concerned with two functions, hearing and body balance.

The outer ear consists of the projecting part (a) _____ (pinna, pharynx) and the passage auditory canal leading to the (b) _____ (thrombus, tympanum). The middle ear contains three bones and a (c) _____ (eustachian tube, effector) which connects the cavity of the middle ear with the throat. The three bones are collectively called the (d) _____ (organelles, ossicles). The inner ear has three parts. One of them is spiral-shaped and looks like a snail shell which is called the (e) _____ (cochlea, canine).

(iii) Choose the **ODD** one out from the following terms and name the category to which the others belong: [5]

- (a) Mitochondria, Nucleotide, Gene, Chromatid
- (b) Urea, Bile, Ammonia, Creatinine
- (c) Hepatic artery, Renal artery, Pulmonary artery, Dorsal aorta
- (d) Oxygen, Hydrogen, Glucose, Starch
- (e) Kidney, Skin, Heart, Lungs

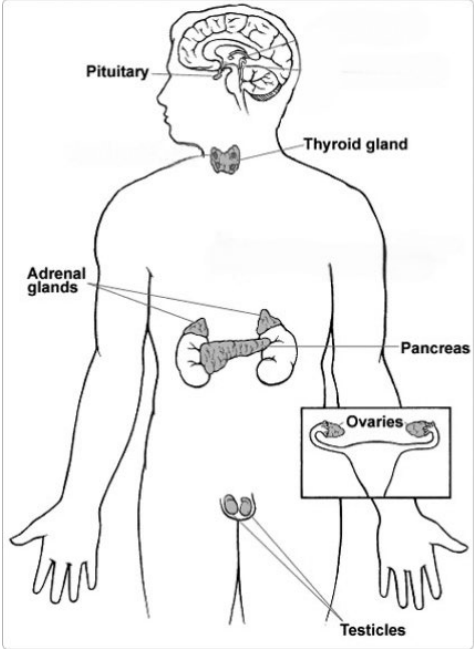
(iv) A potted plant is kept in sunlight and the plant is covered with a polythene cover. Water droplets appear inside the cover after few hours. Answer the following questions: [5]



- (a) Name the process that is observed in the above experiment.
- (b) From which part of the plant does water loss mainly occur?
- (c) How does this process help in water conduction?
- (d) Name *one* environmental factor that can increase the process mentioned in (a).
- (e) What will happen if the process in (a) stops?

- (v) Study the diagram given below and match the gland with its functions: [5]

Example: Thyroid gland – (f) secretes thyroxine

Structure	Functions
 The diagram illustrates the human endocrine system. It shows a human figure with labels for the following glands: Pituitary (located in the brain), Thyroid gland (in the neck), Adrenal glands (two glands, one on each kidney), Pancreas (in the abdominal region), Ovaries (two glands in the female pelvic region, shown in a separate inset box), and Testicles (two glands in the male pelvic region, shown in a separate inset box).	(a) Secretes testosterone (b) Secretes glucagon (c) Secretes oestrogen (d) Secretes vasopressin (e) Secretes adrenaline (f) Secretes thyroxine

SECTION B (40 Marks)

(Attempt **any four** questions from this **Section**.)

Question 3

- (i) Define *mutation*. [1]
- (ii) Give *one* difference between the number of chromosomes passed on to each daughter cell in mitosis and meiosis. [2]
- (iii) Hema, a 25-year-old woman has normal vision, but is a carrier of colour blindness (X^oX). Her husband is colour blind (X^oY). [2]
- (a) What percentage of their sons will be colour blind?
- (b) Can a daughter be colour blind in this case?
- (iv) Mention the functions of the following: [2]
- (a) Scrotum
- (b) Epididymis
- (v) Ratan, a 10-year-old boy was going home from school on a rainy day. He hears a loud thunderbolt and closes his ears suddenly. [3]
- (a) What kind of reflex action is this?
- (b) Which part of the brain is involved in this action?
- (c) Which sense organ triggered it?

Question 4

- (i) State Mendel's law of dominance. [1]
- (ii) Expand the biological abbreviations: [2]
 - (a) TSH
 - (b) CNG
- (iii) Differentiate between an artery and a vein with reference to size of lumen. [2]
- (iv) We urinate fewer times in summer than in winter and the urine is generally thicker. Explain with reasons. [2]
- (v) Draw a neat, labelled diagram of the cell organelle which is responsible for photosynthesis and label the following parts: [3]
 - (a) Stroma
 - (b) Grana

Question 5

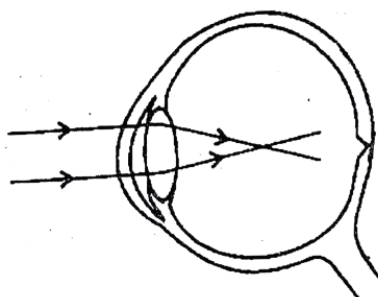
- (i) What is *Parturition*? [1]
- (ii) Sanjay, a 14-year-old boy observed droplets of water on the grass when he went for a walk in the morning. [2]
 - (a) What is this phenomenon of formation of water droplets on grass called?
 - (b) Name the structures through which the water droplets come out.
- (iii) Write *any two* objectives of 'Swachh Bharat Abhiyan'. [2]

(iv) Mention the function of the following plant hormones: [2]

(a) Absciscic acid

(b) Gibberellins

(v) Given below is a diagram depicting a defect of the human eye. Study the same and answer the questions that follow: [3]



(a) Name the defect shown in the diagram.

(b) What is the possible reason that causes this defect?

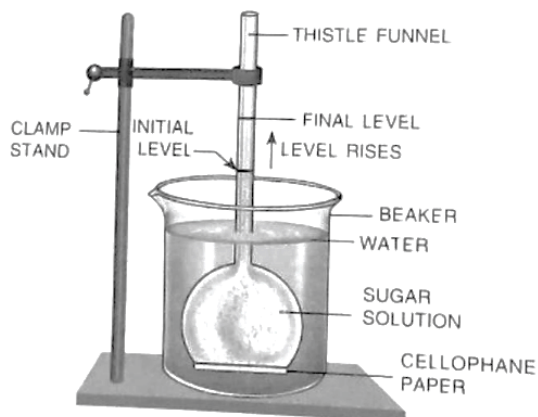
(c) Name the type of lens used to correct this defect.

Question 6

(i) What do you understand by the term *birth rate* in population? [1]

(ii) State *two* differences between Fraternal twins and Identical twins. [2]

- (iii) The experiment below shows the thistle funnel containing sugar solution. [2]



Define the phenomenon taking place in the above experiment.

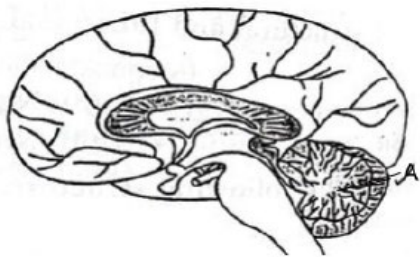
- (iv) What would happen if the cellophane was replaced by the following in the above experiment? [2]
- (a) A rubber sheet
 - (b) A muslin cloth
- (v) Homozygous pea plant bearing red coloured flowers (R) is crossed with a homozygous pea plant bearing white flowers (r): [3]
- (a) Give the phenotype of the F₁ generation.
 - (b) Write the monohybrid ratio of the offsprings of the F₂ generation.
 - (c) What is the genotype obtained in F₂ generation?

Question 7

- (i) Explain the term *Fertilization*. [1]
- (ii) Nitin, a student of class 10, learnt that sperms produced in the testes move into a highly coiled tube and stays there temporarily. [2]
- (a) Name this highly coiled tube.
- (b) What is the function of this coiled tube?
- (iii) Name *any two* vestigial organs found in humans. [2]
- (iv) Arrange the following food chains in a proper sequence beginning with the term that is underlined: [2]
- (a) Corn, Eagle, Fox, Rat
- (b) Nectar, Crow, Bee, Lizard
- (v) Draw a neat, labelled diagram to show Anaphase stage of mitosis in an animal cell having 4 chromosomes. [3]

Question 8

- (i) Mention the number of Cranial nerves in a man. [1]
- (ii) Study the diagram given below and answer the questions that follow: [2]



- (a) Identify the part labelled A.
- (b) What is its function?

- (iii) Mention the *two* main processes in urine formation. [2]
- (iv) Give the exact location of: [2]
- (a) Amniotic fluid
 - (b) Fovea centralis
- (v) People living in congested cities often complain of breathing problems. [3]
Visibility is also reduced due to smog.
- (a) What is the type of pollution that is being referred to?
 - (b) What is *smog*?
 - (c) Name *one* harmful gas released from motor vehicles.